

CONFIDENTIAL

TEST 1

SEMESTER 2, 2014/2015

SUBJECT CODE : **SCSR2043**

SUBJECT : **OPERATING SYSTEMS**

YEARS / COURSE : **SCSJ/SCSR/SCSV/SCSD/SCSB**

TIME : **1 HOUR 30 MINUTES**

DATE : **29 MARCH 2015**

VENUE : **DK7-N24**

INSTRUCTIONS :

Answer **ALL** questions from **SECTION A** and **SECTION B** in question booklet.

(Please fill in your particulars)

Name	
I/C No.	
Years / Course	
Section	
Lecturer's Name	

This questions paper consists of (9) printed pages, including this page.

SECTION A**OBJECTIVE QUESTIONS (15 MARKS)**

(INSTRUCTION: Please answer all questions in the question booklet)

1. Portion of operating system that is in main memory that contains most frequently used functions is known as a _____.
A. interrupt C. supervisor
B. kernel D. manager
2. Which one of the following items is **NOT** part of type of hardware protection?
A. CPU Protection C. Memory controller
B. Memory Protection D. I/O Protection
3. A system call in the operating system will be executed in the _____ mode.
A. manager C. memory
B. user D. kernel
4. Powerful microcomputers developed for use by commercial, educational and government enterprises are called _____.
A. supercomputers C. terminals
B. minicomputers D. workstations
5. _____ systems are computers placed inside other products to add features and capabilities.
A. Real-time C. Embedded
B. Interactive D. Hybrid
6. _____ is a computer that has large physical size and huge amount of internal memory capacity used by government agencies and universities.
A. minicomputer C. mainframe
B. supercomputer D. workstation
7. _____ introduced the need for control cards, which defined the exact nature of each program and its requirements.
A. Job scheduling C. Job control
B. Control scheduling D. Structure control
8. _____ is the partitioning of a single machine, each of which can support a different operating system.
A. Multiprocessing C. Virtualization
B. Multithreading D. Shared processing
9. The primary distinguishing characteristic of modern personal computers is _____.
A. memory capacity and disk space
B. processor and memory capacity
C. disk space and physical size
D. processor capacity and physical size

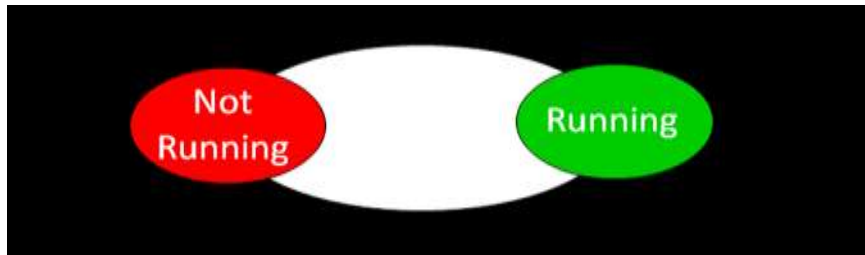
10. _____ systems are used in time-critical environments where reliability is important and data must be processed within a strict time limit.

- A. Embedded
- B. Hybrid
- C. Interactive
- D. Real-time

11. The following are components of a process **EXCEPT**:

- A. information table
- B. program code
- C. context of a program execution
- D. program data

12. The following process state diagram below shows which model?



- A. Queue Model
- B. State Truth Model
- C. Five State Model
- D. Two State Model

13. The following are contents of Process Control Block (PCB) **EXCEPT**:

- A. process creation
- B. process identification
- C. processor state information
- D. process control information

14. The following UNIX system call creates a new process.

- A. main ()
- B. fork ()
- C. exec ()
- D. exit ()

15. The following are characteristics of _____.

- Changes in information when CPU is switched from one process to another.
- Store the current process₀ state for the process that leaves CPU in its PCB₀.
- Load the state of a process₁ that is being allocated CPU from its PCB₁.

- A. Process Scheduler
- B. Overhead
- C. Context Switching
- D. Five State Model

OBJECTIVE ANSWER SHEET

Name	
I/C No.	
Lecturer (Circle)	DR SUBARIAH, RASHIDAH, HAZINAH, FIROZ, MAZURA, MUHALIM, AHMAD FARIZ, DR ISMAIL FAUZI, NOH, ZURIAHATI

SECTION A

Mark your answer clearly.

Example: =A= ☒ =C= =D=

1. =A= =B= =C= =D=

2. =A= =B= =C= =D=

3. =A= =B= =C= =D=

4. =A= =B= =C= =D=

5. =A= =B= =C= =D=

6. =A= =B= =C= =D=

7. =A= =B= =C= =D=

8. =A= =B= =C= =D=

9. =A= =B= =C= =D=

10. =A= =B= =C= =D=

11. =A= =B= =C= =D=

12. =A= =B= =C= =D=

13. =A= =B= =C= =D=

14. =A= =B= =C= =D=

15. =A= =B= =C= =D=

SECTION B**STRUCTURED QUESTIONS (45 MARKS)**

(INSTRUCTION: Please answer all questions in the question booklet)

QUESTION 1 (15 MARKS)

a). List **FIVE** essential subsystem managers of modern operating systems. [5 marks]

b). State the subsystem manager associated with the devices or components of operating systems listed in the table below. [3 marks]

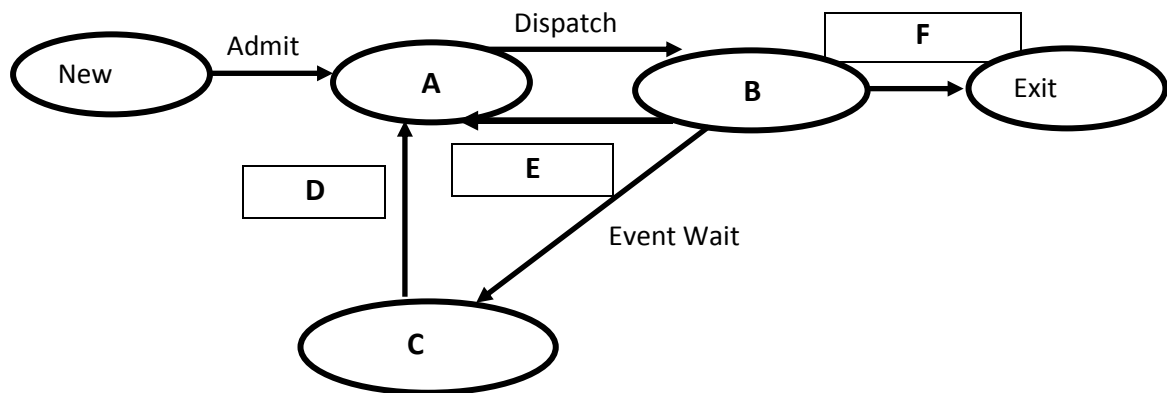
Devices / Components	Answer
Printer, Hard disk, Monitor	
MS Word document files, C++ source code files	
Random access memory	

c). Describe a mechanism for enforcing hardware protection in order to prevent a program from modifying the memory associated with other programs. [5 marks]

d). States **TWO** usage of supercomputers. [2 marks]

QUESTION 2 (10 MARKS)

a). Complete label **A to F** for Five State Process model. [3 marks]

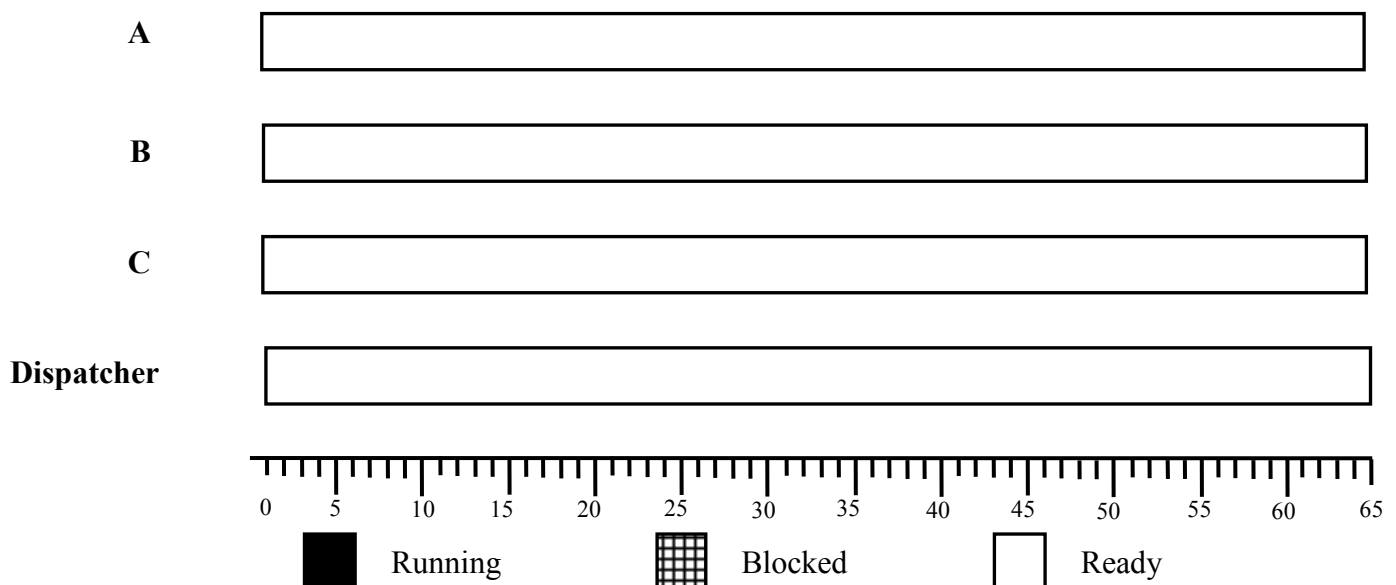


b). The following table below shows the trace addresses for 3 processes A, B, C and the dispatcher in memory. All processes are executed in sequence on a single processor. The execution begins with process B, followed by A and C. Process A request an I/O resource at **instruction cycle 11** and wait up to **20 instruction cycles**. Process C requests an I/O resource at **instruction cycle 33** and wait up to **10 instruction cycles**.

The processes and location.

Process	Addresses
A	1600 - 1609
B	2800 - 2809
C	10000 - 10009
Dispatcher	200 – 203

Complete the figure below by drawing the process state for all processes within **65 instruction cycle** with **THREE** different indicators for **running**, **ready** and **blocked** state. Assume the **CPU time is 4 instruction cycles**. [7 marks]



QUESTION 3 (10 MARKS)

a). Identify the correct description in table below by selecting the answer options given. [3 marks]

Answer Options:

A	child process
B	exit ()
C	shell
D	exec ()

Descriptions	Answer
A command-line interpreter that provides a traditional user interface.	
Will return an integer value of zero (0) to variable pid.	
To terminate a process.	

b). Study the C program given below. Please draw a complete process tree for execution of the program. For each of the process drawn, it must have the **id value, pid1, pid2 and expected output** (as in the example shown). Assume the initial process **id = 5**. [7 marks]

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    int pid1, pid2;

    printf("Hello World \n");
    pid1 = fork();
    if (pid1 < 0)
        printf("Bye Bye World \n");
    else if (pid1 == 0) {
        pid2 = fork();
        printf("I'm the child \n"); }
    else {
        pid2 = fork();
        printf("I'm the parent or another child \n");}
    return 0;
}
```

id= 5
pid1 = pid2 =
Output:

Answer:

QUESTION 4 (10 MARKS)

Identify the correct commands in table below by selecting the answer options given. Assume you are in your working directory. [10 marks]

Answer Options:

A	cat fork1.c
B	gcc fork1.c
C	ls ../
D	mkdir Result
E	rm -r chappie
F	pwd
G	cp fork1.c fork2.c
H	./a.out
I	rmdir Bee3
J	mv fork1.c ../chappie

Tasks	Answer
Execute the output file generated in the directory (default).	
Create a directory with name Result	
List all directories created under the parent of current working directory	
Removes chappie and any files it contains	
View the content of file fork1.c without using the editor	
Compiling fork1.c	
Removing directory Bee3.	
Make a copy of file fork1.c in the same directory with the name fork2.c	
Move the fork1.c file in the current directory into chappie directory.	
To display your current directory now.	